Netherlands Forensic Institute
Ministry of Justice and Security

Using Bayesian networks in forensic science

On model structure, probabilities and communication

JURIX
The Foundation for Legal Knowledge Based Systems

EVIR 4 AI

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Netherlands Forensic Institute
Ministry of Justice and Security

- Independent forensic science provider
- Many areas of expertise: DNA, micro traces, digital traces, firearms, forensic statistics, etc.
- Core tasks: case work, R&D, knowledge sharing

Team “evidence evaluation and statistics”

- Evaluation of the strength of evidence (likelihood ratio)
- **Modeling evaluations (with Bayesian networks)**
- Experimental design, sampling, etc.
- Determination of measurement uncertainty (chemical/physical traces)

Evidence evaluation: the likelihood ratio

Observations, e.g. DNA match

Evidential strength

$$LR = \frac{P(E \mid H_1, I)}{P(E \mid H_2, I)}$$

Often the prosecution scenario

Often the defense scenario

The hierarchy of propositions

1. (SUB)-SOURCE LEVEL

2. ACTIVITY LEVEL

3. OFFENSE LEVEL

Required information
Ultimate issue
Influence case circumstances

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Activity level propositions

- **Actor disputed**

H_1: John broke the glass window.

H_2: Steve broke the glass window and John hugged Steve afterwards.

Forensic Science International 278 (2017) 115–124

Contents lists available at ScienceDirect

Forensic Science International

journal homepage: www.elsevier.com/locate/forsciint

Activity level DNA evidence evaluation: On propositions addressing the actor or the activity

Bas Kokshoorn[a,*,1], Bart J. Blankers[a,1], Jacob de Zoete[a,b], Charles E.H. Berger[a,c]

- **Activity disputed**

H_3: John had sexual intercourse with the complainant and had dinner with the complainant.

H_4: John did not have sexual intercourse with the complainant, John had dinner with the complainant.

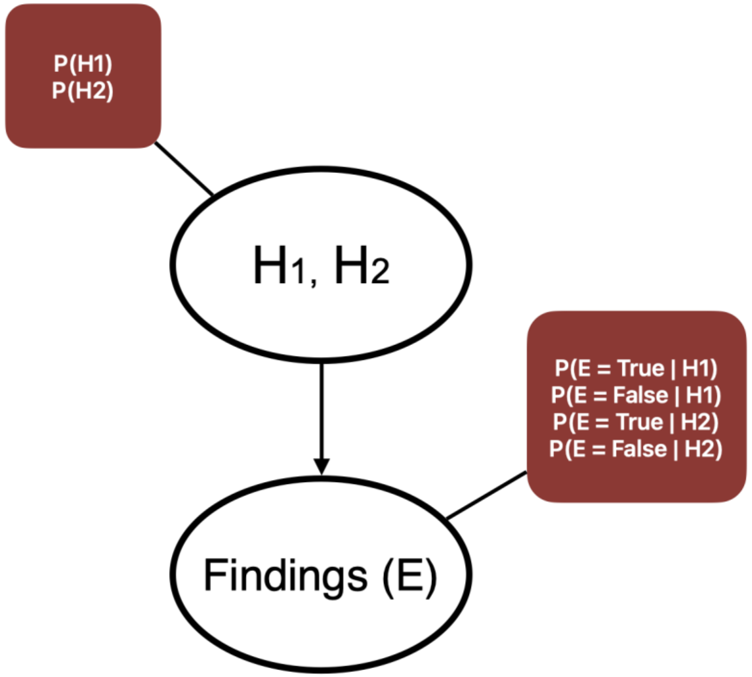
Some challenges

More uncertainties

Missing data

Making complex reasoning understandable and transparent

Graphical representation of Bayes' theorem



- LR framework
- Transparent
- Flexible
- Communication

Using template Bayesian networks

A generic Bayesian network structure that:

- can be reused and adapted for similar types of cases;
- makes network construction easier and more consistent.

For example Taylor et al.'s (2018) template model.

Forensic Science International: Genetics 33 (2018) 136–146

Contents lists available at ScienceDirect

Forensic Science International: Genetics

journal homepage: www.elsevier.com/locate/fsigen

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Research paper

A template for constructing Bayesian networks in forensic biology cases when considering activity level propositions

Duncan Taylor[a,b,*], Alex Biedermann[c], Tacha Hicks[c,d], Christophe Champod[c]

Template Bayesian network

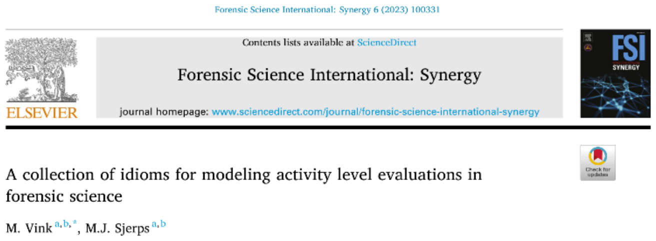
Hypotheses at activity level

Activities that compose the hypotheses

Transfer, persistence, prevalence, recovery of traces given the activities

Group trace that may arise from multiple activities

Group case findings



Forensic Science International: Synergy 6 (2023) 100331

Contents lists available at ScienceDirect

Forensic Science International: Synergy

journal homepage: www.sciencedirect.com/journal/forensic-science-international-synergy

A collection of idioms for modeling activity level evaluations in forensic science

M. Vink[a,b,*], M.J. Sjerps[a,b]

Transfer probabilities

- Probability of DNA transfer, persistence and recovery through stabbing or cooking
- Assign: literature, case-specific experiments, expert knowledge and elicitation
- Test robustness: sensitivity analyses
- Options when LR is ‘sensitive’:
  - Report lower/upper bound
  - Report that a robust LR cannot be determined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Forensic Science International
Volume 355, February 2024, 111944
FSI

A practical treatment of sensitivity analyses in activity level evaluations

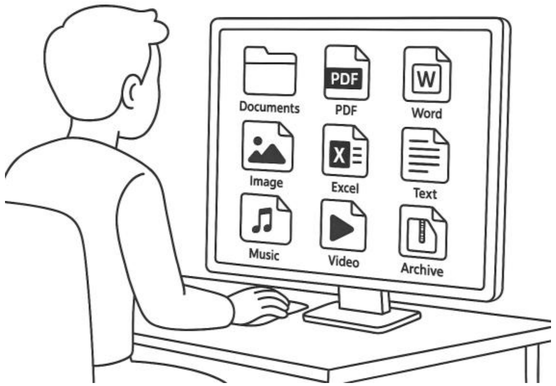
Duncan Taylor [a b], Bas Kokshoorn [c d], Christophe Champod [e]

What about probabilities that are even more ‘belief’ than ‘hard statistics’?

Case example

Case of suspected possession of illegal content on an electronic device.

I was hacked! Someone else controled my computer remotely.

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Forensic Science International: Digital Investigation

Volume 55, December 2025, 302023

FSI

Evaluating digital forensic findings in Trojan horse defense cases using Bayesian networks

Marouschka Vink [a b], Ruud Schramp [b], Bas Kokshoorn [b c], Marjan J. Sjerps [a b]

Case propositions and findings

PROPOSITIONS

H_1: Mr. X downloaded the files to the computer

H_2: Someone else with remote access downloaded the files to the computer. Mr. X is unaware of the presence of the files.

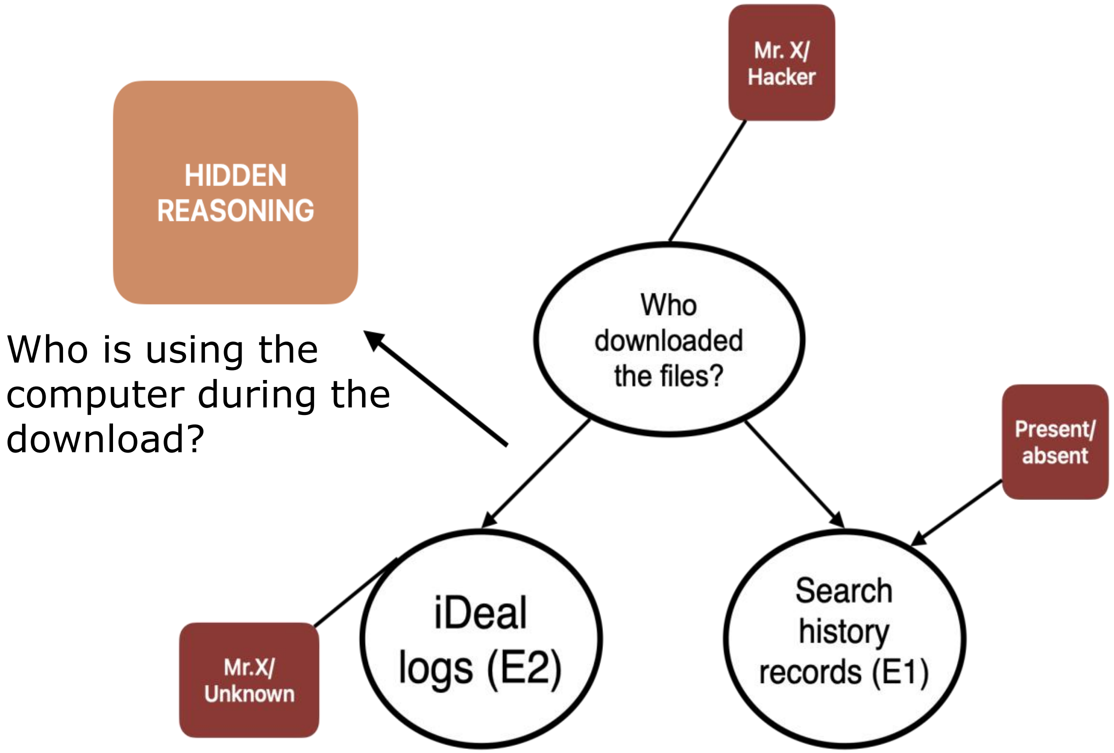
EXAMPLE FINDINGS

Search history records containing terms matching the file names of interest (*Finding 1)*

Bank account logs related to Mr. X on the computer, around the time of the download *(Finding 2)*

For more information on how to formulate propositions in Trojan horse defense cases:
Vink, M., Schramp, R., Berger, C.E.H., Sjerps, M.J., 2025. Formulating propositions in Trojan horse defense cases. Forensic Science International: Digital Investigation 53, 301915.

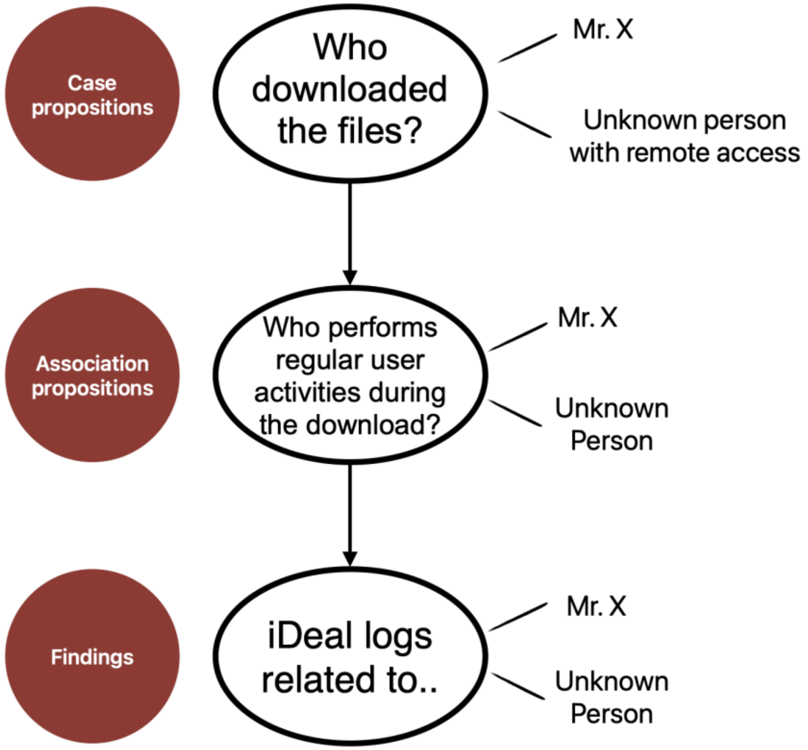
Bayesian network fragment



(Conditional) probabilities

Regular user activities performed by..	Mr. X downloaded the files	Unknown person U downloaded the files
Mr. X	q	*r*
Unknown	1 – q	1 - r

SUBJECTIVE?



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Open debate

- Probability of activities given a scenario, social studies / previous casework.
- More expert belief than hard statistics.
- Who is best to assign such probabilities? The forensic expert?

Final remarks

Bayesian networks:

- are very suitable for activity level evaluations;
- provide a way to communicate “Bayes” to other professionals;
- force to reason about probabilities;
- provide room for discussion;
- are not the only way to graphically represent evidence evaluations.

Thank you!

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For questions and/or
more information:
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